Problem 5

- Solve $y^{(4)} - y = e^x$.
- Consider the HDE: $y^{(4)} - y = 0$.
- Let $y = e^{rx}$ be the solution of HDE and AE is $r^4 - 1 = 0$.
- Then, the roots of AE are $1, -1, \pm i$.
- Thus, $y_c = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$.
- Since 1 is a simple root, $s = 1$: take $y_p = Axe^x$.
- Computing derivatives and substituting: $y_p'''' - y_p = 4Ae^x$.
- Setting $4A = 1 \Rightarrow A = \frac{1}{4}$.

  $y_p = \frac{x}{4} e^x$
- The GS is given by $y = y_c + y_p$

$$\boxed{y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{x}{4} e^x}$$



Problem 6

- Solve $y^{(4)} - 2y'' + y = e^x$.
- Consider the HDE: $y^{(4)} - 2y'' + y = 0$.
- Let $y = e^{rx}$ be the solution of HDE and AE is $r^4 - 2r^2 + 1 = 0$.
- Then, the roots of AE are $1, -1$ (TWICE).
- Thus, $y_c = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x}$.
- Auxiliary equation $(r^2 - 1)^2 = 0$: roots $r = \pm 1$, each with multiplicity 2. Since 1 has multiplicity 2, $s = 2$: take $y_p = Ax^2e^x$.
- Utilizing it is the given DE, we have $8A = 1$.
- So $A = \frac{1}{8}$ and $y_p = \frac{1}{8}x^2e^x$.

$$\boxed{y = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x} + \frac{1}{8}x^2e^x}$$



Problem 7

- Solve $y^{(4)} - 2y'' + y = xe^x$.
- Consider the HDE: $y^{(4)} - 2y'' + y = 0$.
- Let $y = e^{rx}$ be the solution of HDE and AE is $r^4 - 2r^2 + 1 = 0$.
- Then, the roots of AE are $1, -1$ (TWICE).
- Thus, $y_c = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x}$.
- Same $s = 2$ and x is a polynomial of degree 1 multiplied with e^x, take $y_p = x^2(Ax + B)e^x$.
- Using it in DE, we have $24A = 1$ and $24A + 8B = 0$.
- Solving: $A = \frac{1}{24}$, $B = -\frac{1}{8}$. The GS is given by $y = y_c + y_p$

$$\boxed{y = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x} + \frac{x^2(x-3)}{24}e^x}$$



Problem 8

- Solve $y''' + 3y'' + 3y' + y = x + e^{-x}$.
- Auxiliary equation $(r+1)^3 = 0$: root $r = -1$ with multiplicity 3. Then, $y_c = (c_1 + c_2 x + c_3 x^2)e^{-x}$.
- Let $R(x) = R_1(x) + R_2(x)$ with $R_1(x) = x$ and $R_2(x) = e^{-x}$.
- Since 0 is not a root, $s = 0$: take $y_{p_1} = Ax + B$.
- Substituting: $3A + (Ax + B) = x \Rightarrow A = 1$, $3A + B = 0 \Rightarrow B = -3$.

  $y_{p_1} = x - 3$
- For $r = -1$, multiplicity $s = 3$: take $y_{p_2} = Ax^3 e^{-x}$.
- Solving it, we have $y_{p_2} = \frac{x^3}{6} e^{-x}$.
- Hence $y_p = y_{p_1} + y_{p_2} = x - 3 + \frac{x^3}{6} e^{-x}$
- The GS is given by $y = y_c + y_p$

$$\boxed{y = (c_1 + c_2 x + c_3 x^2)e^{-x} + x - 3 + \frac{x^3}{6} e^{-x}}.$$



Problem 9

- Solve $y^{(4)} + 4y = x$.
- the AE for the HDE is $r^4 + 4 = 0$.
- Roots: $1 \pm i, -1 \pm i$ (all complex, none equal to 0).
- Thus, $y_c = e^x(c_1 \cos x + c_2 \sin x) + e^{-x}(c_3 \cos x + c_4 \sin x)$.
- Note that $s = 0$: take $y_p = Ax + B$.
- Substituting: $y_p'''' = 0$, so $4(Ax + B) = x$.
- Matching: $4A = 1 \Rightarrow A = \frac{1}{4}$; $4B = 0 \Rightarrow B = 0$.

  $y_p = \dfrac{x}{4}$
- The GS is given by $y = y_c + y_p$

$$\boxed{y = e^x(c_1 \cos x + c_2 \sin x) + e^{-x}(c_3 \cos x + c_4 \sin x) + \frac{x}{4}}$$



Variation of parameters for a 3rd order ODE

- Consider the third order linear non-homogeneous ODE (in standard form) $y''' + p(x)y'' + q(x)y' + r(x)y = f(x)$.
- Assume the complementary function $y_c = C_1y_1 + C_2y_2 + C_3y_3$ is already known.
- Here y_1, y_2, y_3 are three linearly independent solutions of the homogeneous equation : Wronskian $W(y_1, y_2, y_3) \neq 0$ on domain.
- We seek a particular solution of the form

$$y_p = u_1(x)y_1 + u_2(x)y_2 + u_3(x)y_3$$

- Differentiating y_p directly gives many extra terms involving u_1', u_2', u_3'.
- To simplify, we impose two auxiliary conditions as we differentiate.
- These conditions are chosen to eliminate second derivatives of u_1, u_2, u_3.



Imposing the Auxiliary Conditions

- Differentiate $y_p = u_1y_1 + u_2y_2 + u_3y_3$ once:

$$y_p' = (u_1'y_1 + u_2'y_2 + u_3'y_3) + (u_1y_1' + u_2y_2' + u_3y_3')$$

- **Condition 1:** set $u_1'y_1 + u_2'y_2 + u_3'y_3 = 0$.
- This leaves $y_p' = u_1y_1' + u_2y_2' + u_3y_3'$, just like the homogeneous case.
- Differentiate again:

$$y_p'' = (u_1'y_1' + u_2'y_2' + u_3'y_3') + (u_1y_1'' + u_2y_2'' + u_3y_3'')$$

- **Condition 2:** set $u_1'y_1' + u_2'y_2' + u_3'y_3' = 0$.
- This leaves $y_p'' = u_1y_1'' + u_2y_2'' + u_3y_3''$.
- Differentiate once more to get y_p''', which will include u_1', u_2', u_3' naturally.
- Substituting y_p, y_p', y_p'', y_p''' into the ODE gives the third equation.
- These three equations together form a linear system for u_1', u_2', u_3'.



The Linear System for u_1', u_2', u_3'

- Differentiating y_p'' once more and substituting into the ODE, the homogeneous parts cancel as y_1, y_2, y_3 each individually satisfy the homogeneous DE.
- What remains is exactly $u_1'y_1'' + u_2'y_2'' + u_3'y_3'' = f(x)$.
- Combined with the two earlier conditions, we obtain the 3×3 linear system

$$u_1'y_1 + u_2'y_2 + u_3'y_3 = 0$$

$$u_1'y_1' + u_2'y_2' + u_3'y_3' = 0$$

$$u_1'y_1'' + u_2'y_2'' + u_3'y_3'' = f(x)$$

- The coefficient matrix of this system is exactly the Wronskian matrix of y_1, y_2, y_3.
- Since $W(y_1, y_2, y_3) \neq 0$, this system has a unique solution for u_1', u_2', u_3' at each x.
- We solve it using Cramer's rule.



Solving by Cramer's Rule

- Let $W = W(y_1, y_2, y_3)$ be the Wronskian determinant of the system.
- Let W_1, W_2, W_3 be W with the 1st, 2nd, 3rd column respectively replaced by $(0, 0, f(x))^T$.
- Cramer's rule gives the unique solution

$$u_1' = \frac{W_1}{W}, \qquad u_2' = \frac{W_2}{W}, \qquad u_3' = \frac{W_3}{W}$$

- Each W_i is computed by expanding along the replaced column, since only the last entry is nonzero.
- Integrating each u_i' (dropping constants of integration) gives $u_1(x), u_2(x), u_3(x)$ by leaving the constants of integration.
- The particular solution is then

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

.

- The GS of the non-homogeneous ODE is $y = y_c + y_p$.



Problem 1

- Solve $y''' + y' = \sec x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$
- Characteristic equation: $r^3 + r = 0 \implies r(r^2 + 1) = 0$
- Roots: $r = 0,\ r = \pm i$
- Complementary function: $y_c = C_1 + C_2 \cos x + C_3 \sin x$
- So $y_1 = 1,\ y_2 = \cos x,\ y_3 = \sin x$, and $f(x) = \sec x$
- Compute derivatives: $y_1' = 0, y_1'' = 0$;
  $y_2' = -\sin x, y_2'' = -\cos x$; $y_3' = \cos x, y_3'' = -\sin x$
- Wronskian:

$$W = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix} = 1(\sin^2 x + \cos^2 x) = 1$$

- Since $W = 1$ is constant and nonzero, the setup is ready for Cramer's rule
- Now, we have to compute u_1', u_2', u_3'.



Problem 1: Computing u_1', u_2', u_3'

- Replace column 1 with $(0, 0, \sec x)$ and expand:
  $W_1 = \sec x \cdot (\cos^2 x + \sin^2 x) = \sec x.$
- So $u_1' = W_1/W = \sec x \implies u_1 = \ln|\sec x + \tan x|.$
- Replace column 2 with $(0, 0, \sec x)$ and expand:
  $W_2 = -\sec x \cos x = -1.$
- So $u_2' = W_2/W = -1 \implies u_2 = -x.$
- Replace column 3 with $(0, 0, \sec x)$ and expand:
  $W_3 = -\sec x \sin x = -\tan x.$
- So $u_3' = W_3/W = -\tan x \implies u_3 = \ln|\cos x|.$
- Thus, $y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 =$
  $\ln|\sec x + \tan x| + (-x)\cos x + \sin x \ln|\cos x|.$
- General solution:

$$y = C_1 + C_2 \cos x + C_3 \sin x + \ln|\sec x + \tan x| - x \cos x + \sin x \ln|\cos x|$$



Problem 2:

- Solve $y''' - 3y'' + 3y' - y = \dfrac{e^x}{x^2}$ for $x > 0$
- Characteristic equation:
  $r^3 - 3r^2 + 3r - 1 = 0 \implies (r-1)^3 = 0.$
- Triple root: $r = 1, 1, 1.$
- Complementary function: $y_c = (C_1 + C_2 x + C_3 x^2)e^x.$
- So $y_1 = e^x,\ y_2 = xe^x,\ y_3 = x^2 e^x$, and $f(x) = e^x/x^2$
- For repeated roots, the standard Wronskian formula gives $W = 2e^{3x}$.
- This can be verified directly by expanding the 3×3 determinant of y_1, y_2, y_3 and derivatives.
- Since $W = 2e^{3x} \neq 0$ for all x, Cramer's rule applies smoothly.
- We have to compute u_1', u_2', u_3'.



Problem 2: Computing u_1', u_2', u_3'

- Replace column 1 with $(0, 0, f)$ and expand: $W_1 = f \cdot x^2 e^{2x} = e^{3x}$.
- So $u_1' = W_1/W = \dfrac{e^{3x}}{2e^{3x}} = \dfrac{1}{2} \implies u_1 = \dfrac{x}{2}$.
- Replace column 2 with $(0, 0, f)$ and expand: $W_2 = -f \cdot 2xe^{2x} = -\dfrac{2e^{3x}}{x}$.
- So $u_2' = W_2/W = -\dfrac{1}{x} \implies u_2 = -\ln x$.
- Replace column 3 with $(0, 0, f)$ and expand: $W_3 = f \cdot e^{2x} = \dfrac{e^{3x}}{x^2}$.
- So $u_3' = W_3/W = \dfrac{1}{2x^2} \implies u_3 = -\dfrac{1}{2x}$.
- We now assemble $y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$.



Problem 2: Final Solution

- Assemble the particular solution:

$$y_p = \frac{x}{2}e^x + (-\ln x)(xe^x) + \left(-\frac{1}{2x}\right)(x^2e^x).$$

- Simplify:

$$y_p = \frac{x}{2}e^x - xe^x \ln x - \frac{x}{2}e^x = -xe^x \ln x.$$

- Combine with the complementary function $y_c = (C_1 + C_2x + C_3x^2)e^x$, the general solution is

$$y = (C_1 + C_2x + C_3x^2)e^x - xe^x \ln x.$$



Cauchy–Euler Equations

- The Cauchy–Euler equation is a special linear ODE with variable coefficients.
- It is also called the **equidimensional equation** because of its scaling structure.
- Unlike general variable-coefficient ODEs, it can be solved in closed form.
- No power series or Frobenius machinery is required to solve it.
- It arises in problems with radial or cylindrical symmetry, e.g. Laplace's equation in polar form.
- It also appears in beam deflection, electric circuits, and economics (elasticity models).
- The point $x = 0$ is always a regular singular point of this equation (?)
- Because of its special structure, a simple trial solution $y = x^m$ works directly.
- We will study both the homogeneous and non-homogeneous versions.



Standard Form of the Cauchy–Euler Equation

- The second order homogeneous Cauchy–Euler equation has the form
$$ax^2y'' + bxy' + cy = 0, \qquad a \neq 0.$$
- Here a, b, c are real constants and $x > 0$ (or $x < 0$ by symmetry).
- The distinguishing feature: the power of x matches the order of differentiation.
- That is, x^2 multiplies y'', and x^1 multiplies y'. This exact matching is what makes the equation "equidimensional"
- Dividing by x^2 gives standard form $y'' + \frac{b}{ax}y' + \frac{c}{ax^2}y = 0$.
- Comparing with $y'' + P(x)y' + Q(x)y = 0$: here $P(x) = \frac{b}{ax}$, $Q(x) = \frac{c}{ax^2}$.
- Both $xP(x)$ and $x^2Q(x)$ are constants, hence analytic at $x = 0$.
- So $x = 0$ is always a regular singular point of a Cauchy–Euler equation



Why $y = x^m$ is the Right Trial Solution

- In the equation $ax^2y'' + bxy' + cy = 0$, each term scales the same way under $x \to \lambda x$.
- This scale invariance suggests a solution that is itself a pure power of x.
- Try $y = x^m$ for some constant m (real or complex) to be determined.
- Then $y' = mx^{m-1}$ and $y'' = m(m-1)x^{m-2}$.
- Substituting into $ax^2y'' + bxy' + cy = 0$:

$$ax^2 \cdot m(m-1)x^{m-2} + bx \cdot mx^{m-1} + cx^m = 0.$$

- Simplify powers of x: every term reduces exactly to x^m

$$\big[a\,m(m-1) + b\,m + c\big]x^m = 0.$$

- Since $x^m \neq 0$ for $x > 0$, the bracket must vanish.
- This gives a purely algebraic equation for m, with no series required. It is caled the **auxiliary equation** (or indicial equation) of Cauchy–Euler equation.



The Auxiliary Equation

- The auxiliary equation is

$$a\,m(m-1) + b\,m + c = 0 \implies am^2 + (b-a)m + c = 0.$$

- It has two roots m_1, m_2, found using the quadratic formula

$$m = \frac{-(b-a) \pm \sqrt{(b-a)^2 - 4ac}}{2a}$$

- The nature of these roots depends on the discriminant $(b-a)^2 - 4ac$
- Three cases arise, exactly parallel to solving a constant-coefficient ODE
- **Case 1:** distinct real roots $m_1 \neq m_2$
- **Case 2:** repeated real root $m_1 = m_2$
- **Case 3:** complex conjugate roots $m = \alpha \pm i\beta$
- Each case produces a different form of the general solution



Case 1: Distinct Real Roots

- Suppose the auxiliary equation $am(m-1)+bm+c=0$ has two distinct real roots $m_1 \neq m_2$.
- Then $y_1 = x^{m_1}$ and $y_2 = x^{m_2}$ are both solutions of the ODE.
- Their Wronskian $W(y_1, y_2) = (m_2 - m_1)x^{m_1+m_2-1} \neq 0$ for $x > 0$.
- A nonzero Wronskian confirms y_1, y_2 are linearly independent.
- General solution:

$$y = C_1 x^{m_1} + C_2 x^{m_2}.$$

- This is directly analogous to $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ for constant-coefficient ODEs.
- Here powers of x play the role that exponentials play in the constant-coefficient case.
- If m_1, m_2 are both negative, solutions blow up as $x \to 0^+$.
- If m_1, m_2 are both positive, solutions vanish as $x \to 0^+$.
- This case is the most common and the easiest to handle.



Case 2: Repeated Real Root

- Suppose the auxiliary equation has a repeated root $m_1 = m_2 = m$.
- This happens exactly when the discriminant $(b-a)^2 - 4ac = 0$.
- One solution is immediate: $y_1 = x^m$.
- For the 2nd solution, use the reduction of order $y_2(x) = u(x)x^m$ gives the 2nd solution as $y_2 = x^m \ln x$. (Homework)
- This logarithmic solution is the direct analogue of $y_2 = xe^{rx}$ in constant-coefficient theory.
- General solution is $y = C_1 x^m + C_2 x^m \ln x$.
- Equivalently: $y = x^m(C_1 + C_2 \ln x)$.
- The Wronskian of y_1, y_2 is $W = x^{2m-1} \neq 0$, confirming independence.
- As $x \to 0^+$, the term $x^m \ln x \to 0$ if $m > 0$, but this needs care for $m \leq 0$.



Case 3: Complex Conjugate Roots

- Suppose the auxiliary equation has complex conjugate roots $m = \alpha \pm i\beta$, $\beta \neq 0$.
- Formally, two solutions are $y_1 = x^{\alpha+i\beta}$ and $y_2 = x^{\alpha-i\beta}$.
- Write $x^{i\beta} = e^{i\beta \ln x} = \cos(\beta \ln x) + i \sin(\beta \ln x)$ using $x^{i\beta} = e^{(\ln x)(i\beta)}$.
- This uses the identity $x^{i\beta} = e^{i\beta \ln x}$, valid for $x > 0$.
- Taking real and imaginary parts of $x^{\alpha+i\beta}$ gives two real solutions.
- Real solution 1: $y_1 = x^{\alpha} \cos(\beta \ln x)$.
- Real solution 2: $y_2 = x^{\alpha} \sin(\beta \ln x)$.
- General solution:

$$y = x^{\alpha}\left[C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x)\right]$$

.

- This is the exact analogue of $y = e^{\alpha x}[C_1 \cos \beta x + C_2 \sin \beta x]$.
- Here $\ln x$ plays the role that x plays in the constant-coefficient trigonometric solution.

